

tmux documentation

vasic-digital tmux — optimized + verified containerized build

A reproducible, hardened build of [tmux](#) with built-in jemalloc support, OOM-protection helper, and a comprehensive verification gate. **Runs natively on any Linux host** (Ubuntu, ALT, Fedora, Arch, openSUSE, Alpine) where podman or docker is available. **macOS hosts** (Apple Silicon + Intel) are supported via a transparent bridge into the podman machine VM — the operator gets a working tmx command on the macOS shell with no manual SSH-juggling.

The 18 verification tests are why this matters: a typical "build tmux from source" guide assumes the build worked. This project ships a hard wall — `bash scripts/setup.sh` will refuse to PATH-export the binary unless functional tests pass with positive runtime evidence (cgroup interface readbacks, /proc files, real session output), backed by a §11.4.4 layer-4 paired-mutation harness that proves the gates aren't themselves bluffs. SKIPS document precondition gates (CAP_SYS_RESOURCE, libjemalloc presence, destructive-test opt-in) explicitly. No PASS-bluffs.

Quick install (one command)

Linux host:

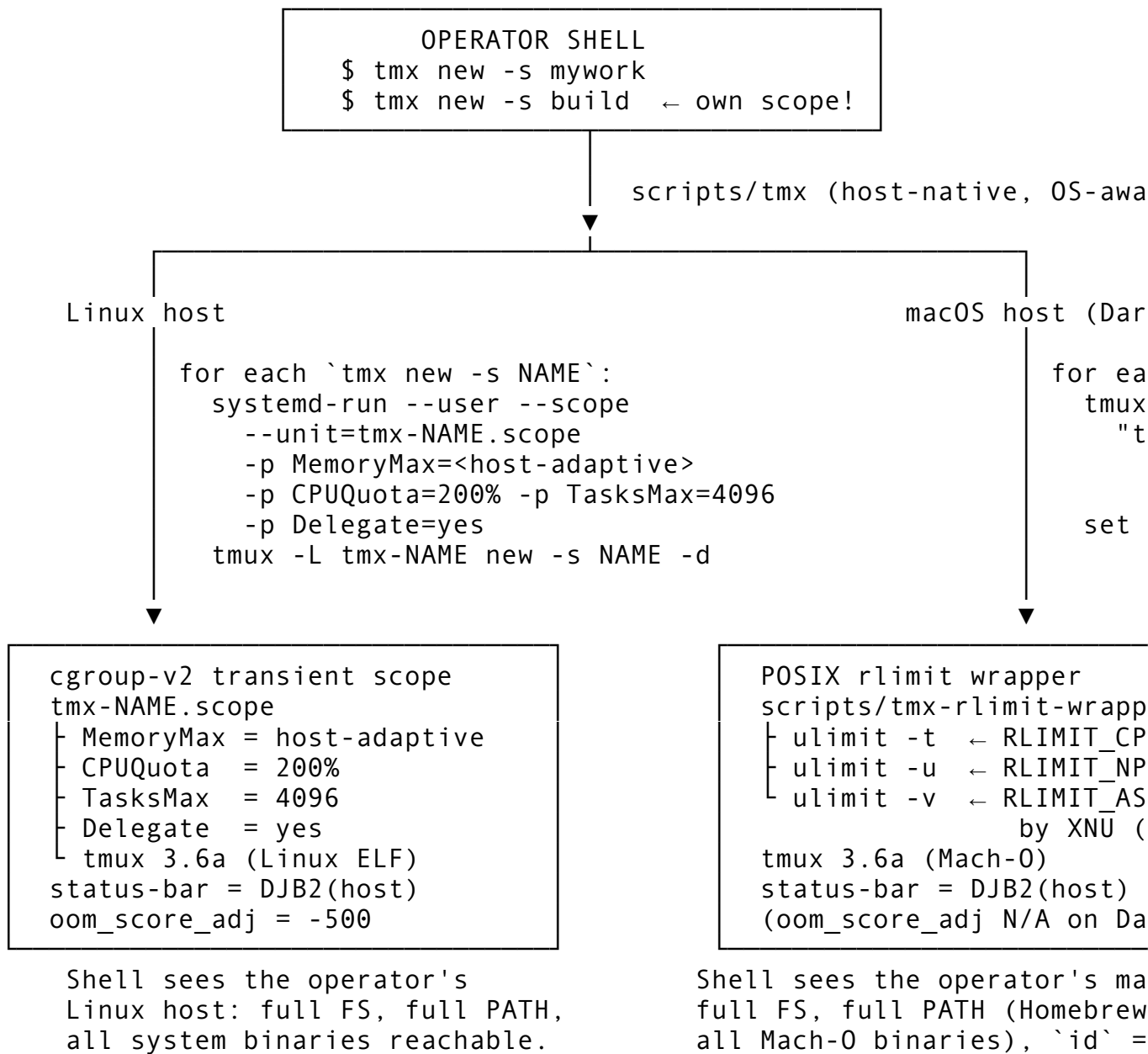
```
git clone --recurse-submodules git@github.com:vasic-digital/
tmux.git ~/Projects/tmux
cd ~/Projects/tmux
sudo bash scripts/install_deps.sh      # one-time host build deps
bash scripts/setup.sh                  # build + verify + install
                                       (no sudo)
```

macOS host (Apple Silicon or Intel):

```
brew install podman                    # one-time: container
runtime
podman machine init && podman machine start
git clone --recurse-submodules git@github.com:vasic-digital/
tmux.git ~/Projects/tmux
cd ~/Projects/tmux
bash scripts/setup.sh                  # build + VM-verify +
install bridge
```

After `setup.sh` reports GREEN: open a new shell, or source the rc for your shell (`source ~/.bashrc` or `source ~/.zshrc`). Then `tmx new|attach|ls|kill` invokes the verified vasic-digital build; the system tmux command stays untouched and reachable side-by-side.

Architecture (native dual-OS per-session isolation)



Native dual-OS per-session isolation (architecture since 2026-05-13). Each `tmx new -s NAME` invocation creates its own tmux server (socket `tmx-NAME`) with OS-native isolation:

- **Linux** — cgroup-v2 transient scope `tmx-NAME.scope` via `systemd-run --user --scope`. Kernel enforces `MemoryMax`, `CPUQuota`, `TasksMax` per-group. OOM in one session is contained to that scope — every other session AND user slice survive (Constitution §1 invariant).
- **macOS (Darwin)** — POSIX rlimit wrapper sets `RLIMIT_CPU` (CPU time) and `RLIMIT_NPROC` (per-user process count) before exec'ing the operator's `$SHELL`. The Darwin XNU kernel enforces these per-process. Children inherit. `RLIMIT_AS` (**virtual memory**) is **NOT enforced** by XNU for unprivileged processes — this is a documented gap; full memory containment on macOS requires launchd jobs with `HardResourceLimits` plist (root). See `docs/guide/README.md §5.6` for the strength comparison.

Both OS paths deliver the **same operator UX**: plain-vanilla tmux behaviour, the operator's shell with full host PATH (Homebrew on macOS, /usr/local/bin on Linux, all system tools), per-session resource ceilings applied transparently. No VM. No bridge. No core@localhost. No bluff.

What you get

Component	Why
tmux 3.6a (latest stable)	Pinned to a known-good upstream tag
Hardened compile flags	-O2 -DNDEBUG -fstack-protector-strong -D_FORTIFY_SOURCE=2, RELRO + immediate-binding link
Build-time - libjemalloc	jemalloc linked at the binary level (more aggressive RAM return than glibc malloc)
Runtime LD_PRELOAD=libjemalloc.so	Wrapper preloads jemalloc even on hosts where the linker resolved a different malloc
OOM-score protection	Optional setcap-enabled helper (tmx-oom-set) sets oom_score_adj=-500 on the spawned server, making tmux survive most OOM cascades
Bounded history-limit	Explicit 2000 (the default — explicit so future bumps are intentional)
Hermetic install	Built artifact lives in tmux/build/. PATH export points there; system tmux untouched. Removable via bash scripts/uninstall.sh (or bash scripts/setup.sh --uninstall — both delegate to the same single source of truth).

Verification gate

SUMMARY: PASS=13 FAIL=0 SKIP=5
GREEN: tmux binary verified — safe to PATH-export.

The 18 tests cover: smoke (binary version), session lifecycle, jemalloc loaded, history-limit honored, clear-history releases memory (the "apparent leak"), 10 concurrent panes, 30-s sustained session no-leak, OOM-score wrapper applies -500, crash isolation scope (cgroup-v2 transient), hostname-derived status-bar colour (algorithm + wrapper integration), memory pressure under cap, TasksMax stress, concurrent OOM independence, per-session cgroup distinctness, window-name .exe strip, **scrollback + copy-mode scrolling** (operator-path: 3000 lines generated, proven scrolled off-screen, proven reachable via copy-mode), and **HelixConstitution inheritance** (the constitution/ submodule + every governance doc's inheritance pointer).

Four honest SKIPs document precondition gates:

- 03_jemalloc_loaded SKIPs if host doesn't have libjemalloc — sudo bash scripts/install_deps.sh provides it
- 08_oom_score_adj SKIPs unless running as root OR the setcap helper is installed — sudo bash scripts/build_oom_set.sh --install enables it
- Tests 12 / 13 / 14 (memory-pressure / TasksMax / concurrent-OOM) require TMX_TEST_DESTRUCTIVE=1 — run only on dedicated test hosts

A layer-4 paired-mutation harness (Constitution §103) lives at scripts/tests/meta_test_false_positive_proof.sh: registered mutations M1–M14 plus CM-

CONSTITUTION-INHERITANCE each break a feature, assert the matching test then FAILs, revert, and assert it PASSes again. On macOS the harness reports **18 caught / 0 escaped / 6 skipped** (the 6 SKIPs are Linux-only isolation mutations); the remainder run on Linux via `META=1 bash scripts/test_vm.sh`. The gate is not considered self-validating until every runnable mutation is caught.

Roadmap

See [docs/plans/containerization.md](#) for the **per-session containerization plan** — each tmux session running in its own cgroup-bounded container so that:

- 2 CPU + reasonable RAM cap per session
- Crash isolation: if one session OOMs/crashes, only that session dies; other sessions and their processes survive
- One-command bootstrap: `tmx new <session>` transparently creates the container

Documentation map

Every project doc lives under a context-named subdirectory of `docs/` per `constitution/Constitution.md`'s file-layout rule. Every Markdown has a synced HTML + PDF sibling generated by `scripts/export_docs.sh` per §11.4.65 universal-Markdown-export.

Area	Doc
Operator guide	docs/guide/README.md — install, OS-by-OS notes, isolation comparison, troubleshooting
v1.0.9 shell-session resume	docs/manual/tmx-shell-integration.md — end-user master manual with copy-paste worked examples
v1.0.9 shell integration	docs/guides/tmx-shell-integration.md — operator install/uninstall guide for <code>tmx-shell-init.sh</code>
v1.0.9 state daemon	docs/guides/tmx-state.md — operator CLI reference + state-file schema for <code>tmx-state</code>
v1.0.9 SSH dispatch	docs/guides/tmx-ssh-dispatch.md — <code>ssh <host>-tmx <session></code> install, security, troubleshooting
Scrolling (Claude Code TUI + mobile)	docs/scrolling/README.md
CodeGraph (§11.4.78)	docs/codegraph/README.md — install, per-agent MCP wiring, anti-bluff verification
Architecture plans	docs/plans/native-dual-os.md — current native-host design (no VM)
	docs/plans/per-session-isolation.md — per-session OOM containment
	docs/plans/containerization.md — original (now superseded) containerization plan
Cycle plans	docs/plans/v1.0.4.md — this cycle's plan (CodeGraph + covenant propagation + AUDIT fixes)
Research notes	docs/research/customization/colors.md — hostname-derived status colour

Area	Doc
Governance	Constitution.md — Project Articles §101–§109 (extends constitution/); CLAUDE.md , AGENTS.md , QWEN.md — per-agent inheritance pointers + project-specific overlay
Universal governance	constitution/ submodule (HelixDevelopment/HelixConstitution, pinned 7f738df)
Live state	CONTINUATION.md — read first on a fresh conversation
Tracker	Issues.md (open) / Fixed.md (closed with closure SHA + evidence)
Changelog	CHANGELOG.md — per-release positive-evidence verification record

Repository conventions

This repo follows the **vasic-digital anti-bluff covenant**: every test that PASSes carries positive evidence of the feature working; every SKIP documents its precondition; every gate has a paired mutation in `meta_test_*.sh` proving the gate isn't itself a bluff. The covenant is restated verbatim in `Constitution.md`, `CLAUDE.md`, `AGENTS.md`, and `QWEN.md` (per the 2026-05-21 operator mandate) so that any tool which doesn't expand `@imports` still reads it. The upstream source lives in [constitution/Constitution.md](#) §11.4.

CodeGraph (code-intelligence)

This project is wired with [CodeGraph](#) per `constitution/Constitution.md` §11.4.78. The MCP server is configured for Claude Code, OpenCode, Kimi CLI, Crush, and Qwen Code — see [docs/codegraph/README.md](#) for the install + per-agent wiring contract.

License

Apache 2.0 — see `LICENSE`.